Hydrothermal synthesis of MoS2 shells

"How does hydrothermal time affect MoS2 shell morphology?" *Generated by LeapSpace*

Hydrothermal time **generally drives MoS_2 shells from thin, compact, and poorly developed coatings toward thicker, rougher, and more fully assembled core-shell or flower-like morphologies**, with sufficiently long times often producing **hollow or uniform shell architectures** rather than just a simple thicker film [115] [116] [119] [120] [121] [122].

How the shell changes with time

Hydrothermal time trend	Typical MoS_2 shell morphology	What it implies
Short time	Initial nuclei, smooth or incomplete coating, or small irregular particles [116] [119] [121] [123]	Shell is still in the nucleation stage and has not fully developed
Intermediate time	Rougher surfaces, looser arrangement, forming nanosheets or incipient core-shell structures [116] [120] [121] [123]	Nuclei grow and self-assemble into layered shell fragments
Longer time	Uniform core-shell shells, hollow microspheres, or dense flower-like/nanoflake shells [115] [116] [119] [120] [121] [122]	Shell growth and rearrangement become more complete and structurally ordered

Mechanism

The time effect is explained by **time-dependent nucleation, anisotropic growth, self-assembly, and Ostwald ripening**: early hydrothermal stages form nuclei, later stages let those nuclei gather into nanosheets, then stack, curl, or reorganize into shell-like or hollow structures [116] [119] [120] [121] [123] [124]. In some systems, prolonged reaction time also reduces the number of layers and changes the phase or crystallinity, which further reshapes the shell texture [125] [126] [127].

Practical takeaway

If you want a **uniform MoS_2 shell**, the evidence supports using a **longer but optimized hydrothermal time** rather than a very short reaction, because short times mainly give nuclei or rough partial shells while longer times allow the shell to consolidate into a defined structure [115] [116] [119] [120] [122]. But the direction is **system-dependent**: too much time can also push the structure toward overgrowth, hollowing, or morphology changes beyond the desired shell [116] [121] [122].

Bottom line: hydrothermal time acts like a morphology "development clock" for MoS_2 shells — **short time = incomplete/compact nuclei, medium time = roughening and shell build-up, long time = mature core-shell or hollow/flower-like shells** [115] [116] [119] [121].

Confidence level: **High** — multiple hydrothermal studies directly show time-driven progression from nuclei to rough shells to mature core-shell or hollow MoS_2 morphologies across different systems [115] [116] [119] [120] [121] [122].

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